SQL Project

Healthcare Data Analysis

Using MySQL to analyse hospital data and generate business insights

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Tools: MySQL | SQL

Introduction of the Project

PROJECT OVERVIEW

Healthcare organizations generate large amounts of data, including patient details, doctor information, visit records, treatment costs, and laboratory test results.

Without proper analysis, it can be difficult for hospital management to understand operational trends and make informed decisions.

In this project, I used SQL to analyse healthcare data and generate meaningful insights about patient volume, diagnoses, follow-ups, treatment costs, laboratory activity, and doctor workloads.

PROJECT FOCUS

10 SQL analysis queries

5 Healthcare tables

1 Goal: actionable insights

Project Objectives

WHAT THE ANALYSIS AIMS TO UNDERSTAND

01

Understand patient volume

Measure total patients and yearly patient trends.

02

Study diagnosis patterns

Identify the most frequently recorded diagnoses.

03

Review follow-up activity

Compare monthly follow-up rates.

04

Understand treatment costs

Calculate total and above-average treatment costs.

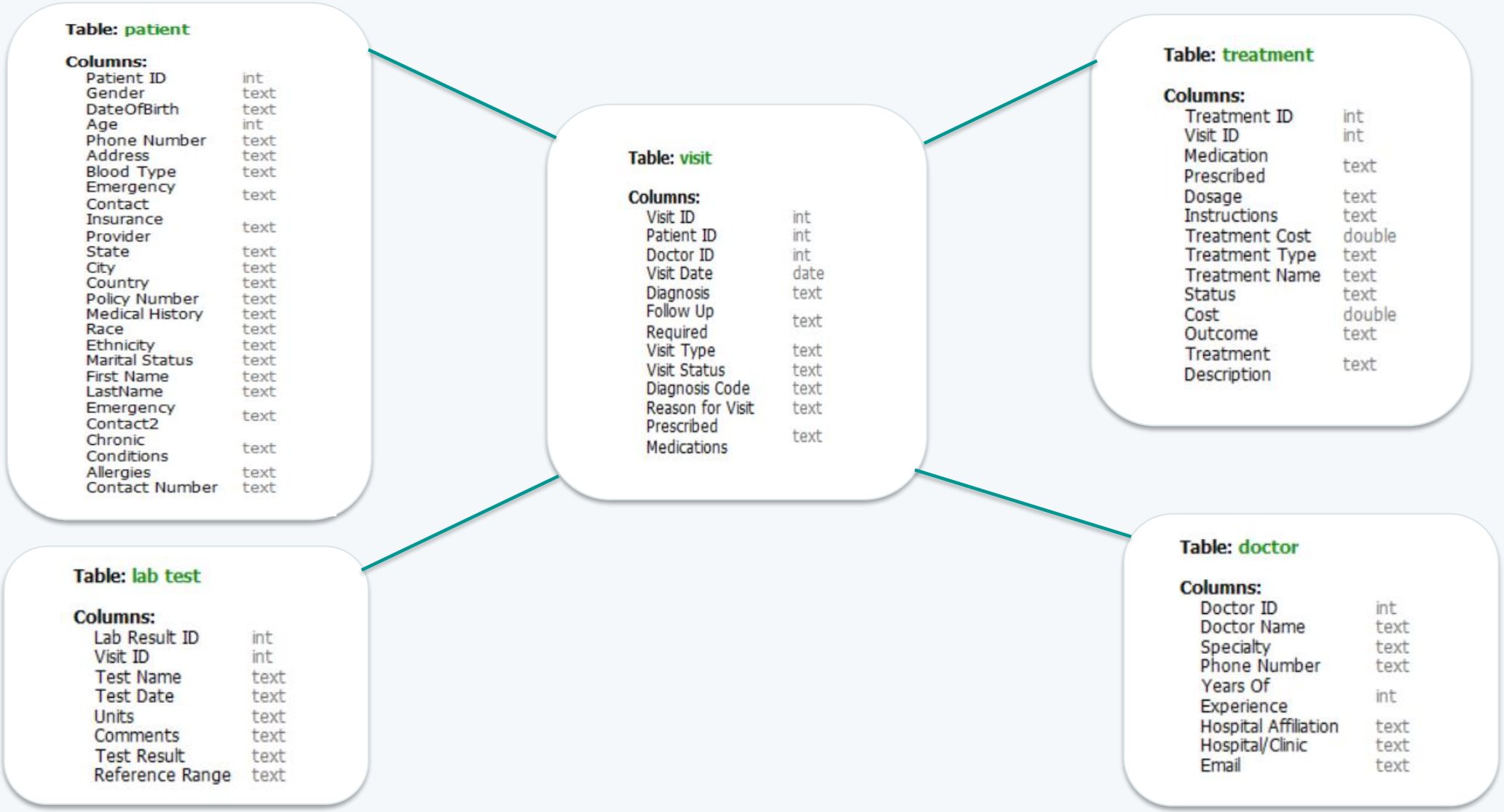
05

Review hospital operations

Analyse lab test volume and doctor workloads.

Schema of the Dataset

MAIN TABLES USED IN THE ANALYSIS



Query 01 — Total Patients

SQL ANALYSIS

How many unique patients are registered in the hospital?

SQL QUERY

```
SELECT
    COUNT(DISTINCT `Patient ID`) AS Total_Patients
FROM patient;
```

WHY DID I CALCULATE IT?

To understand the overall number of unique patients in the hospital.

WHY DID I USE THESE SQL CONCEPTS?

COUNT(DISTINCT) counts each patient ID once and avoids duplicate counting.

RESULT

The hospital has served 10,000 unique patients.

Query 02 — Total Doctors

SQL ANALYSIS

How many unique doctors are available in the hospital?

SQL QUERY

```
SELECT
    COUNT(DISTINCT `Doctor ID`) AS Total_Doctors
FROM doctor;
```

WHY DID I CALCULATE IT?

To understand the size of the available doctor workforce.

WHY DID I USE THESE SQL CONCEPTS?

COUNT(DISTINCT) counts each doctor ID only once.

RESULT

The hospital has 1,000 unique doctors.

Query 03 — Total Treatment Cost

SQL ANALYSIS

What is the total treatment cost recorded in the hospital?

SQL QUERY

```
SELECT
    CONCAT(
        ROUND(SUM(`Treatment Cost`) / 100000, 2), ' L'
    ) AS Total_Treatment_Cost
FROM treatment;
```

WHY DID I CALCULATE IT?

To understand the overall cost associated with treatments.

WHY DID I USE THESE SQL CONCEPTS?

SUM() adds all treatment costs. ROUND() formats the value, and division by 100,000 converts it into lakhs.

RESULT

30.65 lakhs in total treatment cost.

Query 04 — Year-wise Total Patients

SQL ANALYSIS

How many unique patients visited the hospital each year?

SQL QUERY

```
SELECT
    YEAR(v.`Visit Date`) AS Year,
    COUNT(DISTINCT p.`Patient ID`) AS Total_Patients
FROM visit v
JOIN patient p
    ON v.`Patient ID` = p.`Patient ID`
GROUP BY YEAR(v.`Visit Date`)
ORDER BY Year;
```

WHY DID I CALCULATE IT?

To understand how patient volume changes over the years.

WHY DID I USE THESE SQL CONCEPTS?

YEAR() extracts the year. JOIN connects patient and visit records. GROUP BY calculates each year's count.

RESULT

2023: 3,966 | 2024: 4,927 | 2025: 1,107 unique patients.

Query 05 — Top 5 Diagnosed Conditions

SQL ANALYSIS

What are the five most frequently recorded diagnoses?

SQL QUERY

```
SELECT
    Diagnosis,
    COUNT(*) AS Total_Cases
FROM visit
GROUP BY Diagnosis
ORDER BY Total_Cases DESC
LIMIT 5;
```

WHY DID I CALCULATE IT?

To identify the most frequently recorded diagnosis categories.

WHY DID I USE THESE SQL CONCEPTS?

COUNT(*) counts visit records. GROUP BY groups diagnoses, ORDER BY sorts by count, and LIMIT returns five rows.

RESULT

Migraine: 2,039 | Hypertension: 2,011 | Asthma: 2,009
Healthy: 1,981 | Diabetes: 1,960.

Query 06 — Month-wise Follow-up Rate

SQL ANALYSIS

What percentage of visits required follow-up in each month?

SQL QUERY

```
SELECT
    MONTHNAME(`Visit Date`) AS Month,
    CONCAT(ROUND(
            COUNT(CASE WHEN `Follow Up Required` = 'Yes' THEN 1 end)
            * 100.0 / COUNT(*), 2), '%') AS Follow_up_Rate
FROM visit
GROUP BY MONTH(`Visit Date`),MONTH
ORDER BY MONTH(`Visit Date`);
```

WHY DID I CALCULATE IT?

To understand monthly patterns in visits where follow-up was required.

WHY DID I USE THESE SQL CONCEPTS?

CASE WHEN identifies 'Yes' records. COUNT() counts them, and the percentage is calculated against all visits.

RESULT

April had the highest follow-up rate (52.26%). August had the lowest (45.66%).

Query 07 — Monthly Lab Tests

SQL ANALYSIS

How many laboratory tests were recorded in each month?

SQL QUERY

```
SELECT
    MONTHNAME(`Test Date`) AS Month,
    COUNT(`Lab Result ID`) AS Total_Lab_Tests
FROM `lab test`
GROUP BY MONTH(`Test Date`),MONTH
ORDER BY MONTH(`Test Date`);
```

WHY DID I CALCULATE IT?

To understand the monthly volume of laboratory tests and laboratory workload.

WHY DID I USE THESE SQL CONCEPTS?

COUNT() counts lab test records. GROUP BY groups them by month, and ORDER BY sorts the months in calendar order.

RESULT

May recorded the highest number of lab tests (882), while February recorded the lowest (724).

Query 08 — Above Average Treatment Cost

SQL ANALYSIS

Which treatment records have a cost higher than the average treatment cost?

SQL QUERY

```
SELECT `Treatment Type`, `Treatment Cost`
FROM treatment
WHERE `Treatment Cost` > (
    SELECT AVG(`Treatment Cost`)
    FROM treatment);
```

WHY DID I CALCULATE IT?

To identify treatment records with costs above the overall average.

WHY DID I USE THESE SQL CONCEPTS?

AVG() calculates the average cost. The subquery provides that average to the outer query for comparison.

RESULT

The result shows several treatment records with costs higher than the overall average.

Query 09 — Average Patients per Doctor

SQL ANALYSIS

What is the average number of unique patients per doctor?

SQL QUERY

```
WITH Doctor_Patients AS (
    SELECT `Doctor ID`,
        COUNT(DISTINCT `Patient ID`) AS Total_Patients
    FROM visit
    GROUP BY `Doctor ID`)
SELECT
    ROUND(AVG(Total_Patients), 2) AS Avg_Patients_Per_Doctor
FROM Doctor_Patients;
```

WHY DID I CALCULATE IT?

To understand the average patient workload across doctors who had visits.

WHY DID I USE THESE SQL CONCEPTS?

The CTE first calculates unique patients for each doctor. AVG() then calculates the average of those doctor-level counts.

RESULT

The result shows that, on average, each doctor handles 10 patients.

Query 10 — Doctor Performance Ranking

SQL ANALYSIS

How can doctors be ranked based on their total number of visits?

SQL QUERY

```
SELECT
    d.`Doctor ID`,
    COUNT(v.`Visit ID`) AS Total_Visits,
    RANK() OVER (
        ORDER BY COUNT(v.`Visit ID`) DESC
    ) AS Doctor_Rank
FROM doctor d
LEFT JOIN visit v
    ON d.`Doctor ID` = v.`Doctor ID`
GROUP BY d.`Doctor ID`
ORDER BY Doctor_Rank;
```

WHY DID I CALCULATE IT?

To compare visit volumes across doctors and review workload distribution.

WHY DID I USE THESE SQL CONCEPTS?

LEFT JOIN keeps all doctors, COUNT() calculates visits, and RANK() ranks doctors from highest to lowest visit count.

RESULT

The result shows doctors ranked based on their total number of patient visits.

Key Insights

FINDINGS FROM THE SQL ANALYSIS

Patient volume: The patient table contains 1,000 unique patients.

Yearly trend: Unique patients were 3,966 in 2023, 4,927 in 2024, and 1,107 in 2025.

Diagnosis patterns: Migraine, hypertension, and asthma were the three highest-count diagnosis categories.

Follow-up pattern: April had the highest follow-up rate at 52.26%, while August had the lowest at 45.66%.

Treatment costs: The total recorded treatment cost was 30.65 lakhs.

Recommendations

HOW THE INSIGHTS MAY SUPPORT HOSPITAL OPERATIONS

Resource planning

Use patient volume and yearly trends to support staffing and capacity planning.

Doctor workload

Review patient and visit distribution across doctors to identify workload differences.

Follow-up management

Review monthly follow-up patterns and plan follow-up services accordingly.

Laboratory operations

Use monthly lab test volumes to plan laboratory staff and equipment.

Cost monitoring

Review higher-cost treatment records to better understand treatment-related expenses.

Conclusion

Overall, this project helped me understand how SQL can be used to analyse healthcare data and identify important business insights.

These insights can help hospital management improve resource planning, manage doctors' workloads more efficiently, and make better operational decisions.

THANK YOU